CORE PURE / COMPLEX NUMBERS

De Moivre's Theorem

Mark scheme 13 questions · 115 marks

EXAM BOARDS Edexcel 9FM0 · AQA 7367 · OCR A H245 · OCR MEI H645 · CIE 9231

Questions

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	Total	**115**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes. A grey line gives the rule where the name alone doesn't.

ft	follow through Correct work from the student's own earlier wrong value earns the mark.	**AG**	answer given The answer is printed in the question, so every step to it must be shown.
cao	correct answer only No follow-through from an earlier error.	**cso**	correct solution only No error anywhere in the part, not just in its answer.
oe	or equivalent	**awrt**	answers which round to
soi	seen or implied Earned if written, or if later working shows it was used.	**isw**	ignore subsequent working A correct answer, once reached, isn't spoiled by later working.
SC	special case Credit, as the note says, for work outside the main method.	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- [$\boldsymbol{n}$] ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of $\boldsymbol{g}$

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1(a)	$(\cos\theta + i\sin\theta)^3 = \cos 3\theta + i\sin 3\theta$ $= \cos^3\theta + 3i\cos^2\theta\sin\theta - 3\cos\theta\sin^2\theta - i\sin^3\theta$	**M1**	Uses de Moivre's theorem and attempts a binomial expansion with at least the real terms. Condone coefficient or sign slips.
	$\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta$	**A1**	Correct real part equated to $\cos 3\theta$; imaginary terms need not be seen.
	$\cos 3\theta = \cos^3\theta - 3\cos\theta\left(1 - \cos^2\theta\right)$ $= \cos^3\theta - 3\cos\theta + 3\cos^3\theta$	**M1**	Substitutes $\sin^2\theta = 1 - \cos^2\theta$ into their real part and expands to an expression in $\cos\theta$ only.
	$\cos 3\theta = 4\cos^3\theta - 3\cos\theta$	**A1 AG**	Correct real part leading without error to the printed result; combined steps accepted.
		[4]	
1(b)	$\cos 3\theta = \cos 2\theta$	**M1**	Uses (a) to replace the LHS by $\cos 3\theta$.
	$3\theta = 2n\pi \pm 2\theta \Rightarrow \theta = \ldots$	**M1**	Uses a general solution of $\cos A = \cos B$ with both signs, a factor formula, or lists equal-cosine angles over a sufficient range, leading to $\theta = \ldots$.
	$\theta = 2n\pi$ or $\theta = \frac{2n\pi}{5}$	**A1**	Any two of 0, $\frac{2\pi}{5}$, $\frac{4\pi}{5}$, or both correct general families.
	$\theta = 0,\ \frac{2\pi}{5},\ \frac{4\pi}{5}$	**A1**	All three exact values, including 0, and no others in $0 \leq \theta \leq \pi$.
		[4]	
Total for Question 1		**8**	

Question	Scheme	Marks	Notes
2	$\cos 4\theta + i\sin 4\theta = (\cos\theta + i\sin\theta)^4$ $= \cos^4\theta + 4i\cos^3\theta\sin\theta$ $+ 6i^2\cos^2\theta\sin^2\theta + \ldots$	**M1**	Uses de Moivre's theorem with a binomial expansion of $(\cos\theta + i\sin\theta)^4$, at least its real terms with coefficients 1, 6, 1. Powers of i need not be simplified.
	$\cos 4\theta = \cos^4\theta - 6\cos^2\theta\sin^2\theta + \sin^4\theta$	**A1**	Correct real part equated to $\cos 4\theta$, with powers of i simplified.
	$= \cos^4\theta - 6\cos^2\theta(1 - \cos^2\theta)$ $+ (1 - \cos^2\theta)^2$	**M1**	Uses $\sin^2\theta = 1 - \cos^2\theta$ to write their real part in $\cos\theta$ only. Condone sign slips.
	$= \cos^4\theta - 6\cos^2\theta + 6\cos^4\theta$ $+ 1 - 2\cos^2\theta + \cos^4\theta$	**dM1**	Expands the brackets, with $(1 - \cos^2\theta)^2$ as three terms, and collects powers of $\cos\theta$. Condone sign slips.
	$\cos 4\theta = 8\cos^4\theta - 8\cos^2\theta + 1$	**A1**	cso. The expression, or $a = 8$, $b = -8$, $c = 1$ stated.
	Total for Question 2	**5**	

Question	Scheme	Marks	Notes
3	$\left(z-\frac{1}{z}\right)^3 = z^3 - 3z + \frac{3}{z} - \frac{1}{z^3}$	**M1 A1**	**M1** Expands $\left(z-\frac{1}{z}\right)^3$, by the binomial theorem or otherwise, to four terms in powers of z. Condone sign and coefficient slips.
	$\left(z^3 - \frac{1}{z^3}\right) - 3\left(z - \frac{1}{z}\right)$ $= 2i\sin 3\theta - 6i\sin\theta$	**M1**	Pairs z^k with $-z^{-k}$ in their expansion and replaces each pair by $2i\sin k\theta$. Condone coefficient slips.
	$-8i\sin^3\theta = 2i\sin 3\theta - 6i\sin\theta$ $\Rightarrow A = \frac{3}{4},\ B = -\frac{1}{4}$	**A1**	cso, including the correct factor from $(2i)^3 = -8i$. Accept the identity written with these values.
	Total for Question 3	**4**	

Question	Scheme	Marks	Notes
4(a)	$z = \cos\theta + i\sin\theta \Rightarrow z^n + z^{-n} = 2\cos n\theta$ $z + z^{-1} = 2\cos\theta$	**B1**	Stated or used, for general n or for $n = 1$, 3, 5. Accept $z = e^{i\theta}$.
	$(2\cos\theta)^5 = \left(z + z^{-1}\right)^5$ $= z^5 + 5z^3 + 10z + 10z^{-1} + 5z^{-3} + z^{-5}$	**M1 A1**	**M1** Expands $\left(z + z^{-1}\right)^5$ binomially to six terms in powers of z. Condone coefficient slips. **A1** Correct expansion; the factor 32 may appear later.
	$32\cos^5\theta = 2\cos 5\theta + 10\cos 3\theta + 20\cos\theta$	**M1**	Pairs z^k with z^{-k} and replaces each pair by $2\cos k\theta$, for at least two pairs.
	$A = \frac{1}{16},\ B = \frac{5}{16},\ C = \frac{5}{8}$	**A1**	All three correct, oe; accept the identity with these coefficients.
		[5]	
4(b)	$\int_0^{\frac{\pi}{3}} \cos^5\theta\,\mathrm{d}\theta = \left[\frac{1}{80}\sin 5\theta + \frac{5}{48}\sin 3\theta + \frac{5}{8}\sin\theta\right]_0^{\frac{\pi}{3}}$	**M1**	Integrates their identity from (a), each $\cos k\theta$ to a multiple of $\sin k\theta$. Condone coefficient slips.
	$= \frac{1}{80}\sin\frac{5\pi}{3} + \frac{5}{48}\sin\pi + \frac{5}{8}\sin\frac{\pi}{3} - 0$	**dM1**	Substitutes $\frac{\pi}{3}$ into their integral and subtracts the value at 0, which may be implied.
	$= \dfrac{49\sqrt{3}}{160}$	**A1**	Any exact equivalent, cso.
		[3]	
Total for Question 4		**8**	

Question	Scheme	Marks	Notes
5(a)	$(\cos\theta + i\sin\theta)^5 = \cos 5\theta + i\sin 5\theta$ $= \ldots + 5i\cos^4\theta\sin\theta$ $+ 10i^3\cos^2\theta\sin^3\theta + i^5\sin^5\theta$	**M1**	Applies de Moivre and expands binomially, at least the terms giving the imaginary part. Condone slips in coefficients or powers of i.
	$\sin 5\theta = 5\cos^4\theta\sin\theta - 10\cos^2\theta\sin^3\theta + \sin^5\theta$	**A1**	Correct, from equating imaginary parts, oe.
	$\sin 5\theta = 5\sin\theta(1-\sin^2\theta)^2$ $- 10\sin^3\theta(1-\sin^2\theta) + \sin^5\theta$	**M1**	Uses $\cos^2\theta = 1 - \sin^2\theta$ to write their imaginary part in powers of $\sin\theta$ alone.
	$= 5\sin\theta - 10\sin^3\theta + 5\sin^5\theta$ $- 10\sin^3\theta + 10\sin^5\theta + \sin^5\theta$	**dM1**	Expands brackets and collects like powers of $\sin\theta$. Condone arithmetic slips.
	$\sin 5\theta = 16\sin^5\theta - 20\sin^3\theta + 5\sin\theta$	**A1 AG**	Correct imaginary part and substitution leading without error to the printed result; combined expansion and collection accepted.
		[5]	
5(b)	$\sin 5\theta = \sin 2\theta$	**M1**	Uses part (a) to write the equation as $\sin 5\theta = \sin 2\theta$.
	$5\theta = 2\theta + 2k\pi$ or $5\theta = \pi - 2\theta + 2k\pi \Rightarrow \theta = \ldots$	**M1**	Forms both families of solutions for equal sines, oe, or factorises $\sin 5\theta - \sin 2\theta$ into a product and solves each factor. Condone sign slips.
	$\theta = \frac{2k\pi}{3} \Rightarrow \theta = 0,\ \frac{2\pi}{3}$	**A1**	Both 0 and $\frac{2\pi}{3}$.
	$\theta = \frac{(2k+1)\pi}{7} \Rightarrow \theta = \frac{\pi}{7},\ \frac{3\pi}{7},\ \frac{5\pi}{7},\ \pi$	**A1**	All four of $\frac{\pi}{7}$, $\frac{3\pi}{7}$, $\frac{5\pi}{7}$, π.
	$\theta = 0,\ \frac{\pi}{7},\ \frac{3\pi}{7},\ \frac{2\pi}{3},\ \frac{5\pi}{7},\ \pi$	**A1**	All six exact values, including 0 and π, and no others in $0 \leq \theta \leq \pi$.
		[5]	
	Total for Question 5	**10**	

Question	Scheme	Marks	Notes
6(a)	$(z+z^{-1})^6 = 64\cos^6\theta$	**B1**	
	$(z+z^{-1})^6 = z^6+6z^4+15z^2+20$ $+15z^{-2}+6z^{-4}+z^{-6}$	**M1 A1**	**M1** Attempts the binomial expansion: seven terms with binomial coefficients and powers of z. Powers need not be simplified; condone one coefficient slip. **A1** Correct expansion with powers simplified.
	$64\cos^6\theta = (z^6+z^{-6})+6(z^4+z^{-4})+\ldots$ $= 2\cos 6\theta + 12\cos 4\theta + 30\cos 2\theta + 20$	**M1**	Pairs z^n with z^{-n} and replaces by $2\cos n\theta$ for at least two pairs, with their coefficients.
	$a=\frac{1}{32},\ b=\frac{3}{16},\ c=\frac{15}{32},\ d=\frac{5}{16}$	**A1**	All four, oe, stated or in the identity.
		[5]	
6(b)	$\frac{1}{32}\int_0^{\pi/4}(\cos 6\theta + 6\cos 4\theta + 15\cos 2\theta + 10)\,\mathrm{d}\theta$ $= \frac{1}{32}\left[\frac{1}{6}\sin 6\theta + \frac{3}{2}\sin 4\theta + \frac{15}{2}\sin 2\theta + 10\theta\right]_0^{\pi/4}$	**M1 A1**	**M1** Integrates their part (a) identity term by term to the form $p\sin 6\theta + q\sin 4\theta + r\sin 2\theta + s\theta$. Condone sign and coefficient slips. **A1** Correct integral, oe.
	$= \frac{1}{32}\left(-\frac{1}{6}+0+\frac{15}{2}+\frac{5\pi}{2}\right)-0$ $= \frac{11}{48}+\frac{5\pi}{64}$	**dM1** **A1 AG**	**dM1** Substitutes $\theta=\frac{\pi}{4}$ and $\theta=0$ and subtracts; the zero lower limit may be implied. **A1 AG** Correct substituted expression with sine values evaluated, leading without error to the printed result; combined steps accepted.
		[4]	
	Total for Question 6	**9**	

Question	Scheme	Marks	Notes
7(a)	$z^n = \cos n\theta + i\sin n\theta$ $z^{-n} = \cos(-n\theta) + i\sin(-n\theta)$ $= \cos n\theta - i\sin n\theta$	**M1**	Uses de Moivre's theorem to write both z^n and z^{-n} in terms of $\cos n\theta$ and $\sin n\theta$; z^{-n} may come from $\frac{1}{z^n}$ by realising the denominator.
	$z^n + \frac{1}{z^n} = 2\cos n\theta$	**A1**	From correct expressions for both powers, with $\cos(-n\theta)$ and $\sin(-n\theta)$ correctly simplified.
	$z^n - \frac{1}{z^n} = (\cos n\theta + i\sin n\theta) - (\cos n\theta - i\sin n\theta)$ $= 2i\sin n\theta$	**A1 AG**	Correct subtraction of the two expressions leading without error to the printed result.
		[3]	
7(b)	$\left(z + \frac{1}{z}\right)^3 = z^3 + \frac{1}{z^3} + 3\left(z + \frac{1}{z}\right)$ $\Rightarrow 8\cos^3\theta = 2\cos 3\theta + 6\cos\theta$	**M1**	Expands the sum cube binomially, groups into $z^3 + z^{-3}$ and $z + z^{-1}$, and replaces each by (a). Condone sign and coefficient slips.
	$\cos 3\theta = 4\cos^3\theta - 3\cos\theta$	**A1**	Correct, from the sum-cube expansion and (a).
	$\left(z - \frac{1}{z}\right)^3 = z^3 - \frac{1}{z^3} - 3\left(z - \frac{1}{z}\right)$ $\Rightarrow -8i\sin^3\theta = 2i\sin 3\theta - 6i\sin\theta$	**M1**	Expands the difference cube binomially, groups into $z^3 - z^{-3}$ and $z - z^{-1}$, and replaces each by (a). Condone sign and coefficient slips.
	$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$	**A1**	Correct, from the difference-cube expansion and (a), with $(2i\sin\theta)^3 = -8i\sin^3\theta$ correct.
	$\tan 3\theta = \frac{3\sin\theta - 4\sin^3\theta}{4\cos^3\theta - 3\cos\theta}$ $= \frac{3\tan\theta\sec^2\theta - 4\tan^3\theta}{4 - 3\sec^2\theta}$	**M1**	Writes $\tan 3\theta$ as their $\sin 3\theta$ over their $\cos 3\theta$, each a cubic in $\sin\theta$ or $\cos\theta$, and divides through by a power of $\cos\theta$ to introduce $\tan\theta$.
	$= \frac{3\tan\theta(1+\tan^2\theta) - 4\tan^3\theta}{4 - 3(1+\tan^2\theta)}$	**M1**	Attempts to replace $\sec^2\theta$ by $1 + \tan^2\theta$ in their quotient to obtain an expression in $\tan\theta$ alone. Condone algebraic slips; an equivalent conversion using $\sin^2\theta + \cos^2\theta = 1$ is accepted.
	$= \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$	**A1 AG**	From correct expressions for $\sin 3\theta$ and $\cos 3\theta$, with $\sec^2\theta = 1 + \tan^2\theta$ oe used correctly, leading without error to the printed result.
		[7]	
	Total for Question 7	**10**	

Question	Scheme	Marks	Notes
8(a)	$\cos 5\theta + i\sin 5\theta = (\cos\theta + i\sin\theta)^5$ $= \cos^5\theta + 5i\cos^4\theta\sin\theta - 10\cos^3\theta\sin^2\theta + \ldots$	**M1**	Uses de Moivre with a binomial expansion of $(\cos\theta + i\sin\theta)^5$; at least the real terms, with binomial coefficients. Condone slips in powers of i.
	$\cos 5\theta = \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta$	**A1**	Equates real parts correctly.
	$= \cos^5\theta - 10\cos^3\theta(1-\cos^2\theta)$ $+ 5\cos\theta(1-\cos^2\theta)^2$	**M1**	Replaces $\sin^2\theta$ by $1-\cos^2\theta$ in both terms containing $\sin\theta$.
	$= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta$ $+ 5\cos\theta - 10\cos^3\theta + 5\cos^5\theta$	**dM1**	Expands the brackets and collects to a form $a\cos^5\theta + b\cos^3\theta + c\cos\theta$. Condone arithmetic slips.
	$\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$	**A1 AG**	Correct real part, substituted and expanded without error to the printed identity; combined steps accepted.
		[5]	
8(b)	$x = 2\cos\theta \Rightarrow x^5 - 5x^3 + 5x$ $= 32\cos^5\theta - 40\cos^3\theta + 10\cos\theta = 2\cos 5\theta$	**M1**	Substitutes $x = 2\cos\theta$ and uses (a) to write $x^5 - 5x^3 + 5x$ as $2\cos 5\theta$.
	$2\cos 5\theta + 1 = 0 \Rightarrow \cos 5\theta = -\frac{1}{2}$	**A1**	
	$5\theta = \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}, \frac{10\pi}{3}, \frac{14\pi}{3} \Rightarrow \theta = \ldots$	**M1**	Finds at least two values of 5θ for their $\cos 5\theta = k$, $\lvert k\rvert \leq 1$, and divides by 5.
	$x = 2\cos\frac{2\pi}{15}, 2\cos\frac{4\pi}{15}, 2\cos\frac{8\pi}{15}, 2\cos\frac{2\pi}{3}, 2\cos\frac{14\pi}{15}$	**A1**	All five, each as $2\cos q\pi$ oe (e.g. $2\cos\frac{16\pi}{15}$), no repeated root. Decimals, -1, or θ values alone score A0.
		[4]	
	Total for Question 8	**9**	

Question	Scheme	Marks	Notes
9(a)	$(z - z^{-1})^6 = (2i\sin\theta)^6 = -64\sin^6\theta$	**B1**	
	$(z - z^{-1})^6$ $= z^6 - 6z^4 + 15z^2 - 20 + 15z^{-2} - 6z^{-4} + z^{-6}$	**M1 A1**	**M1** Expands using binomial coefficients 1, 6, 15, 20 with powers of z and z^{-1}. Condone sign and power slips; need not be simplified. **A1** All seven terms correct; powers need not be simplified.
	$(z^6 + z^{-6}) - 6(z^4 + z^{-4}) + 15(z^2 + z^{-2}) - 20$ $= 2\cos 6\theta - 12\cos 4\theta + 30\cos 2\theta - 20$	**M1**	Pairs terms and uses $z^n + z^{-n} = 2\cos n\theta$ for at least two pairs. Condone coefficient slips.
	$-64\sin^6\theta = 2\cos 6\theta - 12\cos 4\theta + 30\cos 2\theta - 20$ $\Rightarrow \sin^6\theta = \frac{1}{32}(10 - 15\cos 2\theta + 6\cos 4\theta - \cos 6\theta)$	**A1 AG**	Equates the two expressions and reaches the printed identity without error; combined rearrangement accepted.
		[5]	
9(b)	$x = \tan\theta,\ \mathrm{d}x = \sec^2\theta\,\mathrm{d}\theta$ $\Rightarrow \int \frac{\tan^6\theta}{(\sec^2\theta)^4}\sec^2\theta\,\mathrm{d}\theta$	**M1**	Substitutes fully: $\mathrm{d}x$ replaced by $\sec^2\theta\,\mathrm{d}\theta$ and $1 + x^2$ by $\sec^2\theta$ or $1 + \tan^2\theta$. Condone slips in powers.
	$\int_0^{\pi/4}\sin^6\theta\,\mathrm{d}\theta$	**A1**	
	$\frac{1}{32}\int_0^{\pi/4}(10 - 15\cos 2\theta + 6\cos 4\theta - \cos 6\theta)\,\mathrm{d}\theta$ $= \frac{1}{32}\left[10\theta - \frac{15}{2}\sin 2\theta + \frac{3}{2}\sin 4\theta - \frac{1}{6}\sin 6\theta\right]_0^{\pi/4}$	**M1 A1**	**M1** Replaces $\sin^6\theta$ by the result of (a) and integrates to the form $a\theta + \sum b_k \sin k\theta$. Condone sign and coefficient slips. **A1** Correct integral; the factor $\frac{1}{32}$ may be applied later.
	$\frac{1}{32}\left(\frac{10\pi}{4} - \frac{15}{2} + 0 + \frac{1}{6}\right)$	**dM1**	Substitutes $\frac{\pi}{4}$ and 0 into their integral and subtracts; the zero from the lower limit need not be shown.
	$= \frac{15\pi - 44}{192}$	**A1**	Any exact equivalent, such as $\frac{5\pi}{64} - \frac{11}{48}$.
		[6]	
	Total for Question 9	**11**	

Question	Scheme	Marks	Notes
10(a)	$(\cos\theta + i\sin\theta)^6 = \cos 6\theta + i\sin 6\theta$ $= \cos^6\theta + 6i\cos^5\theta\sin\theta$ $- 15\cos^4\theta\sin^2\theta + \ldots$	**M1**	Equates $\cos 6\theta + i\sin 6\theta$ to a binomial expansion of $(\cos\theta + i\sin\theta)^6$, at least its real terms, with powers of i applied. Condone coefficient and sign slips.
	$\cos 6\theta = \cos^6\theta - 15\cos^4\theta\sin^2\theta$ $+ 15\cos^2\theta\sin^4\theta - \sin^6\theta$	**A1**	
	$u = \sin^2\theta$: $\cos 6\theta = (1-u)^3 - 15(1-u)^2u$ $+ 15(1-u)u^2 - u^3$	**M1**	Replaces $\cos^2\theta$ by $1 - \sin^2\theta$ throughout their real part.
	$= 1 - 18u + 48u^2 - 32u^3$ $= (1-2u)(\ldots)$	**dM1**	Expands and collects into a cubic in $\sin^2\theta$, then factorises it or expands the printed product to compare. Condone slips.
	$\cos 6\theta \equiv (1 - 2\sin^2\theta)(16\sin^4\theta$ $- 16\sin^2\theta + 1)$	**A1 AG**	Correct substituted expression leading without error to the printed identity; matching the expanded printed RHS with a conclusion is accepted.
		[5]	
10(b)	$(1-2u)(16u^2 - 16u + 1) = 0 \Rightarrow u = \ldots$	**M1**	Sets each factor to zero and solves the quadratic in $\sin^2\theta$ by formula with their values or completing the square.
	$u = \frac{1}{2},\ \frac{2\pm\sqrt{3}}{4}$	**A1**	All three, oe such as $\frac{16\pm8\sqrt{3}}{32}$.
	$\cos\left(6\times\frac{\pi}{12}\right) = \cos\frac{\pi}{2} = 0$ $0 < \frac{\pi}{12} < \frac{\pi}{4} \Rightarrow \sin^2\frac{\pi}{12} < \frac{1}{2}$	**M1**	Uses $\theta = \frac{\pi}{12}$ as a solution of $\cos 6\theta = 0$ and gives a reason rejecting $\frac{1}{2}$ and $\frac{2+\sqrt{3}}{4}$, such as $\sin^2\frac{\pi}{12} < \frac{1}{2}$.
	$\sin^2\left(\frac{\pi}{12}\right) = \frac{2-\sqrt{3}}{4}$	**A1 AG**	cso, with both other values rejected.
		[4]	
10(c)	$\sin^2\left(\frac{5\pi}{12}\right) = \frac{2+\sqrt{3}}{4}$	**B1**	Any exact form.
		[1]	
	Total for Question 10	**10**	

Question	Scheme	Marks	Notes
11(i)	$\cos 5\theta + i\sin 5\theta = (\cos\theta + i\sin\theta)^5$ $= \cos^5\theta + 5i\cos^4\theta\sin\theta$ $- 10\cos^3\theta\sin^2\theta + \ldots$	**M1**	Uses de Moivre with a binomial expansion of $(\cos\theta + i\sin\theta)^5$, showing at least the real terms. Condone slips in powers of i.
	$\cos 5\theta = \cos^5\theta - 10\cos^3\theta\sin^2\theta$ $+ 5\cos\theta\sin^4\theta$	**A1**	
	$\cos 5\theta = \cos\theta\big((1-\sin^2\theta)^2$ $- 10(1-\sin^2\theta)\sin^2\theta + 5\sin^4\theta\big)$	**M1**	Replaces $\cos^2\theta$ by $1-\sin^2\theta$ in their real part, keeping a factor $\cos\theta$.
	$= \cos\theta\big(1 - 2\sin^2\theta + \sin^4\theta$ $- 10\sin^2\theta + 15\sin^4\theta\big)$	**dM1**	Expands and collects into powers of $\sin\theta$. Condone arithmetic slips.
	$\cos 5\theta \equiv \cos\theta(1 - 12\sin^2\theta + 16\sin^4\theta)$	**A1 AG**	Correct substitution leading without error to the printed identity; expansion and collection may be combined.
		[5]	
11(ii)	$\frac{\cos 5\theta}{\cos\theta} = 1 - 12u + 16u^2$, $u = \sin^2\theta$	**M1**	Divides the identity by $\cos\theta$ and treats the result as a quadratic in $\sin^2\theta$, any letter.
	$0 \leq u < 1$	**B1**	Strict inequality at 1 needed; equivalent statements in $\sin\theta$ accepted.
	$u = \frac{12}{2\times 16}$ $[= \frac{3}{8}]$	**M1**	Finds the turning point of their quadratic by differentiation, completing the square or $-\frac{b}{2a}$. Condone sign slips.
	$1 - 12\left(\frac{3}{8}\right) + 16\left(\frac{3}{8}\right)^2 = -\frac{5}{4}$	**A1**	$-\frac{5}{4}$ oe.
	$u \to 1 \Rightarrow 1 - 12 + 16 = \ldots$	**M1**	Evaluates their quadratic at, or as u tends to, $u = 1$.
	5, not attained: $u = 1 \Rightarrow \cos\theta = 0$	**A1**	5 with the reason it is not attained. Solving $f(u) = 5$ is not required.
	$-\frac{5}{4} \leq \frac{\cos 5\theta}{\cos\theta} < 5$	**A1**	Correct inequality at both ends; $\left[-\frac{5}{4}, 5\right)$ oe accepted.
		[7]	
	Total for Question 11	**12**	

Question	Scheme	Marks	Notes
12	$z - z^{-1} = 2i\sin\theta,\ z + z^{-1} = 2\cos\theta$ $\Rightarrow \sin\theta\cos^4\theta = \frac{1}{32i}(z - z^{-1})(z + z^{-1})^4$	**B1**	Both, stated or used.
	$(z - z^{-1})(z^4 + 4z^2 + 6 + 4z^{-2} + z^{-4})$	**M1**	Multiplies out $(z - z^{-1})(z + z^{-1})^4$ to a sum of powers of z, in any order of multiplication. Condone coefficient and sign slips.
	$= z^5 + 3z^3 + 2z - 2z^{-1} - 3z^{-3} - z^{-5}$	**A1**	Correct expansion, oe; the factor $\frac{1}{32i}$ need not be shown here.
	$(z^5 - z^{-5}) + 3(z^3 - z^{-3}) + 2(z - z^{-1})$ $= 2i\sin 5\theta + 6i\sin 3\theta + 4i\sin\theta$	**M1**	Pairs terms as $z^n - z^{-n}$ and replaces each by $2i\sin n\theta$, keeping their coefficients.
	$A = \frac{1}{8},\ B = \frac{3}{16},\ C = \frac{1}{16}$	**A1**	cso; withhold if an i or the factor $\frac{1}{32}$ is lost and recovered. Accept the identity written with these values.
	Total for Question 12	**5**	

Question	Scheme	Marks	Notes
13(a)	$(z-z^{-1})^4(z+z^{-1})^2 = (2i\sin\theta)^4(2\cos\theta)^2$ $= 64\sin^4\theta\cos^2\theta$	**B1**	
	$(z-z^{-1})^4(z+z^{-1})^2 = (z^2-2+z^{-2})(z^4-2+z^{-4})$ $= \ldots$	**M1**	Attempts the full expansion of the product into powers of z, in any order. Condone sign and coefficient slips.
	$= z^6 - 2z^4 - z^2 + 4 - z^{-2} - 2z^{-4} + z^{-6}$	**A1**	Correct seven-term expansion, oe.
	$= (z^6 + z^{-6}) - 2(z^4 + z^{-4}) - (z^2 + z^{-2}) + 4$ $= 2\cos 6\theta - 4\cos 4\theta - 2\cos 2\theta + 4$	**M1**	Pairs z^n with z^{-n} and replaces each pair by $2\cos n\theta$.
	$64\sin^4\theta\cos^2\theta = 2\cos 6\theta - 4\cos 4\theta - 2\cos 2\theta + 4$ $\Rightarrow \sin^4\theta\cos^2\theta \equiv \frac{1}{32}(\cos 6\theta - 2\cos 4\theta - \cos 2\theta + 2)$	**A1 AG**	Correct expansion equated to $64\sin^4\theta\cos^2\theta$, leading without error to the printed identity.
	Alternative: Double-angle formulae in $\cos 2\theta$		
	$\sin^4\theta\cos^2\theta = \left(\frac{1-\cos 2\theta}{2}\right)^2\left(\frac{1+\cos 2\theta}{2}\right) = \ldots$	**M1**	Uses $\sin^2\theta = \frac{1}{2}(1-\cos 2\theta)$, $\cos^2\theta = \frac{1}{2}(1+\cos 2\theta)$ and expands to a polynomial in $\cos 2\theta$. Condone sign slips.
	$= \frac{1}{8}(1 - \cos 2\theta - \cos^2 2\theta + \cos^3 2\theta)$	**A1**	
	$(\cos 2\theta + i\sin 2\theta)^3 = \cos 6\theta + i\sin 6\theta$ $\Rightarrow \cos 6\theta = \cos^3 2\theta - 3\cos 2\theta\sin^2 2\theta = \ldots$	**M1**	Expands $(\cos 2\theta + i\sin 2\theta)^3$ and equates real parts to express $\cos 6\theta$ in powers of $\cos 2\theta$; $\cos 4\theta$ may come from a double-angle formula. Not for quoting the triple-angle formula.
	$\cos^2 2\theta = \frac{1}{2}(1 + \cos 4\theta)$ $\cos^3 2\theta = \frac{1}{4}(\cos 6\theta + 3\cos 2\theta)$	**A1**	Both, oe.
	$\sin^4\theta\cos^2\theta =$ $\frac{1}{8}\left(1 - \cos 2\theta - \frac{1+\cos 4\theta}{2} + \frac{\cos 6\theta + 3\cos 2\theta}{4}\right)$ $\equiv \frac{1}{32}(\cos 6\theta - 2\cos 4\theta - \cos 2\theta + 2)$	**A1 AG**	Correct substitution into the correct cubic, leading without error to the printed identity.
		[5]	
13(b)	$\sin^2\theta\cos^4\theta = \sin^4\alpha\cos^2\alpha$ $= \frac{1}{32}(\cos 6\alpha - 2\cos 4\alpha - \cos 2\alpha + 2)$	**M1**	Replaces $\sin\theta$ by $\cos\alpha$ and $\cos\theta$ by $\sin\alpha$ and applies the identity of (a) in α.
	$\cos 2\alpha = -\cos 2\theta,\ \cos 4\alpha = \cos 4\theta,\ \cos 6\alpha = -\cos 6\theta$	**M1**	Uses $n\alpha = \frac{n\pi}{2} - n\theta$ to write all three cosines in terms of θ. Condone sign slips.
	$\sin^2\theta\cos^4\theta \equiv \frac{1}{32}(-\cos 6\theta - 2\cos 4\theta + \cos 2\theta + 2)$	**A1**	oe.
		[3]	

Question 13 continued

Question	Scheme	Marks	Notes
13(c)	$\frac{1}{32}(\cos 6\theta - 2\cos 4\theta - \cos 2\theta + 2)$ $+ \frac{2}{32}(\text{“} - \cos 6\theta - 2\cos 4\theta + \cos 2\theta + 2\text{”})$	**M1**	Adds the identity from (a) to twice their identity from (b), of the form $\frac{1}{32}$(cosines of $2\theta, 4\theta, 6\theta$ and a constant).
	$= \frac{1}{32}(-\cos 6\theta - 6\cos 4\theta + \cos 2\theta + 6)$	**A1**	
	$\frac{1}{32}\left[-\frac{1}{6}\sin 6\theta - \frac{3}{2}\sin 4\theta + \frac{1}{2}\sin 2\theta + 6\theta\right]_0^a$	**M1**	Integrates at least one cosine term to the form $k\sin n\theta$ and substitutes the limits 0 and a.
	$\frac{1}{32}\left(6a + \frac{1}{2}\sin 2a - \frac{3}{2}\sin 4a - \frac{1}{6}\sin 6a\right)$ $= \frac{\pi}{32} - \frac{\sqrt{3}}{64}$	**A1**	Correct expression in a, oe, equated to the given value.
	$I\left(\frac{\pi}{6}\right) = \frac{1}{32}\left(\pi + \frac{\sqrt{3}}{4} - \frac{3\sqrt{3}}{4}\right) = \frac{\pi}{32} - \frac{\sqrt{3}}{64}$	**A1 AG**	Substitutes the stated value and obtains the given integral exactly, without error.
	$I'(a) = \sin^2 a\cos^2 a(\sin^2 a + 2\cos^2 a) > 0$ for $0 < a < \frac{\pi}{2}$, so I is strictly increasing and the solution is unique.	**B1**	Explains that the integrand is positive throughout the interval, so the integral is strictly increasing and no other value can satisfy the condition. A numerical or graphical check alone does not score.
		[6]	
	Total for Question 13	**14**	